

PRICE DATASET

The price dataset only encompasses AAPL stock price. We created this dataset to see how well we can predict stock direction using only historical stock data. It will serve as a baseline dataset for modeling. We cleaned this dataset using similar methods to our cleaning of the combined dataset.

Using what we learned during EDA, there were not any noticeable outliers within the stock price data. A time plot of stock price shows consistent movement throughout the time series without any irregularly high or low prices. The only outliers present in our datasets were related to news sentiment.

We also learned that there were no missing values in the stock price data. There is a recorded closing price for each trading day included in the training data. This is to be expected when working with stock data, especially for a valuable company like Apple.

Nothing was done to remove outliers or impute missing values for the price dataset.

In order to prepare the price dataset for classification models, we created stock price direction lags and stock price direction variables in the same way as the combined dataset. The binary price direction lag variables will serve as predictor variables and the stock price direction variable will be the target.

Since the AAPL stock price dataset was already fairly clean and free of missing values, the most significant cleaning was done to prepare the data for a classification problem by creating target and predictor variables that represent stock directional movement.

SENTIMENT DATASET

Our data preparation process began with a careful exploratory data analysis, which revealed that the most significant challenge in the dataset was the presence of missing news data in the earlier years. These gaps were not random; rather, they reflected periods in which fewer news articles were available or archived. Because natural language processing (NLP) models are highly sensitive to data quality and consistency, it is important to address them as a part of our cleaning strategy. Rather than immediately discarding early years or aggressively imputing artificial content, we preserved temporal integrity while minimizing the introduction of bias. The first step in preparing the news data for modeling involved consolidating the text columns. The dataset originally contained separate title and content columns. Since both fields contribute contextual meaning, we combined them into a single `full_text` column. This approach ensures that the model captures both headline-level signals and the deeper contextual information contained within the article body. After creating this unified text field, we removed columns that were not directly relevant to model construction. This helped streamline the dataset, reduce dimensionality, and prevent unnecessary noise from entering the feature space.

Next, we transformed the dataset from article-level observations into daily aggregates. All news articles published on the same date were grouped into a single row representing that day. The `full_text` entries for each date were concatenated to create one consolidated daily document. At the same time, the sentiment variables, negative, neutral, and positive scores, were aggregated using the mean for each day. After constructing the daily news dataset, we merged it with the stock price data using the daily closing price. This alignment allowed us to directly connect daily news sentiment and textual information with corresponding market outcomes. However, introducing stock data revealed additional missing rows in the merged dataset. These gaps were expected and stemmed from two primary sources: trading days without news releases and earlier periods where news coverage was incomplete or unavailable.

We addressed missing textual data conservatively. We replaced missing `full_text` values with blank strings. In a TF-IDF framework, blank strings simply generate zero-valued vectors, accurately representing the absence of news without injecting noise into the model. For the sentiment variables, we implemented a forward-fill strategy with a seven-day limit. This decision reflects a realistic market assumption: the informational impact of news does not disappear immediately but instead decays over time. By forward-filling sentiment values for up to one week, we allow recent sentiment to carry residual influence while preventing outdated news from exerting long-term artificial effects. Any days beyond this window that still lacked prior sentiment were assigned a value of zero, representing a neutral or information-absent state.

Finally, we constructed our classification target variable, “Stock Price Direction.” For each trading day, the stock was labeled “Up” if the closing price exceeded the previous day’s close and “Down” if it was less than or equal to the prior close. This binary formulation transforms the problem into a supervised classification task, enabling direct comparison between news-driven predictions and price-based historical models.

COMBINED DATASET

The combined dataset encompasses both news article sentiment and AAPL stock price. This dataset was created in Week 2's notebook, and each observation contains the daily average sentiment polarity as well as the closing day stock price.

A small number of outliers were identified using the sentiment polarity timeseries from Week 3's data exploration assignment (attached in the appendix). These outliers had abnormally low sentiment polarities, contrasted with positive sentiment scores on the preceding or following days. When directly compared with the stock price timeseries, however, these negative scores directly coincided with dips in the stock price, especially those in late 2020. These negative sentiment polarities could be very useful to a classification model predicting stock price direction, and as a result, these outliers were elected to stay a part of the dataset.

There was also a moderate amount of missing data, all from the news sentiment dataset. These dates are primarily in 2018 & 2019 and are a result of no articles being published on the AAPL stock that day. One basic assumption we can make is that sentiment will be relatively stable for a short period of time until a new article is published with differing sentiment. This period of time was adjudicated to be approximately a week, given the short memories of the public. A forward fill was then used to fill missing news sentiment data for up to a week, any observations with no preceding data more recent than a week remained with missing data. Those observations will be handled after variable creation.

Another consideration with this dataset was whether to standardize the numeric sentiment polarity scores. These scores are clearly bounded (between -1 and 1) and are centered (0 being perfectly neutral). The current distribution is also very intuitive because a negative number corresponds to a negative sentiment while a positive number corresponds to a positive sentiment. Scores closer to zero represent a more neutral sentiment. Due to the intuitive interpretation of this score and the bounded and centered nature, the variable was not standardized further. This may be reconsidered if we use a neural network model as these models strongly prefer standardized data.

The next step was to create lagged columns for both the stock price and sentiment polarity scores. Knowing the history of both these variables could be extremely helpful in predicting next day stock price direction, so lagged variables were created up to five days in the past. Additionally, the stock price direction variable was created, comparing current day stock price to the previous day stock price. This variable is the target variable, as declared in last week's assignment. Stock price direction lags were also created, but these will be used as feature variables. These lags were encoded to be binary, receiving a 1 for the "Up" direction and 0 for "Down".

Next, all observations with missing observations were removed from the dataset. This was done after the forward fill step, meaning all observations removed during this step had missing lag information and/or the news sentiment data was stale for at least a week. Additionally, all duplicate rows were removed, but this ended up being none since the primary key date ensures all rows are unique.

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The variables for the final dataset were then selected, including “Stock Price Direction”, “Date”, sentiment polarity lags up to five days, and stock price direction lags up to four days. Due to the lack of perfect information, the current day news sentiment polarity score was *not* included. The reasoning behind this is anyone trying to forecast stock price direction would not have access to current day news sentiment as it could arrive after the forecast had already been made.

This logic was wrapped up in a data cleaning function and was applied equally on the training, validation, and test datasets. Each dataset was treated separately ensuring there was no data leakage.

APPENDIX

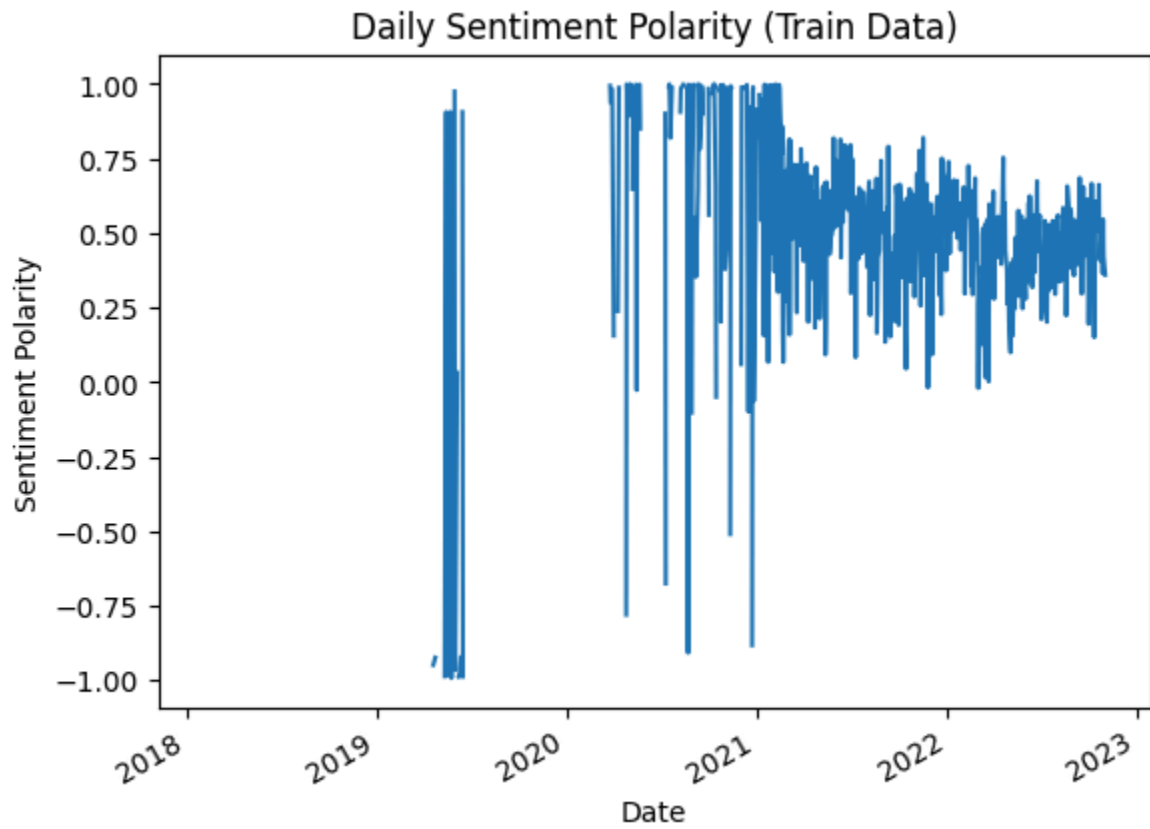


Figure 1. Daily Sentiment Polarity in AAPL News Articles